Ma-sum Abdul-Hamid

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Research Interests

Biosensors and diagnostics, Biomaterials, Microrobotics, AI in Healthcare (RL, CV), and Translational Healthtech Business Strategy.

Education

Doctor of Philosophy: Biomedical Engineering (Diefenderfer fellow) 01/2024 - 12/2027
The Pennsylvania State University **GPA**: 3.97/4.0

Master of Science: Biomedical Engineering 01/2024 - 05/2026
The Pennsylvania State University **GPA**: 3.97/4.0
Relevant Coursework: Cell and Molecular Bioengineering, Systems Physiology, Law and Ethics in Biomedical AI, Numerical Methods for Biomedical Engineers, and Fundamentals of Quantitative Biology.

Bachelor of Science: Mechanical Engineering 09/2019 - 05/2023
New York University Abu Dhabi **GPA**: 3.77/4.0
Relevant Coursework: CAD, Inorganic Chemistry, C++, Probability and Statistics, Linear Algebra, MVC, Differential Equations

Experience

Entrepreneurial Lead/ Venture Co-Lead 12/2024 – Current
Penn State Hemostatic Foam Commercialization Project (medtech startup) – State College, United States of America
- Led NSF I-Corps customer discovery for a bioresorbable chitosan–starch hemostatic foam; secured $55,000 in funding and conducted 100+ stakeholder interviews to refine the value proposition, target market, and go-to-market strategy.
- Developed IP, FDA 510(k), and preclinical validation roadmap under regulatory guidance from Kim Trautman.
- Currently leading bench validation studies for the hemostatic foam, including time-to-hemostasis, uptake capacity, anticoagulated-blood performance, and comparator testing to support preclinical development.
- Secured the Penn State Center for Biodevices Ignite Seed Grant ($60,000) and currently leading pursuit of additional translational funding, including NSF TTP-T ($1.2 million) and NSF SBIR Phase I ($305,000), to advance remaining preclinical, regulatory, and commercialization milestones.

Graduate Teaching Assistant 08/2024 – Current
SP 2026, SP 2025: Numerical Simulation in Biomedical Eng. (BME 401 - Dr. I.S Aronson) - State College, United States of America
- Taught computational modeling and numerical simulation techniques for biomedical applications, emphasizing ethical design considerations and patient-specific problem-solving using COMSOL.

Fall 2024: Analysis of Physiological Systems (BME 301 - Dr. S. Kothapalli) - State College, United States of America
- Taught weekly labs on physiological systems analysis, emphasizing biomedical signal processing in MATLAB for numerical analysis and visualization of biological signals.

Graduate Research Assistant 01/2024 – Current
Active Biomaterials Laboratory (Aronson Lab) – State College, United States of America
- Developed a reduced-order PPO-based control framework for chiral magnetic micropropeller cargo transport, integrating low-Reynolds-number hydrodynamics, magnetic mode switching, cargo capture/release dynamics, and actuation latency; achieved faster autonomous transport than PID-style control in simulation.
- Designed and prototyped an image-activated microfluidic cell sorter with swiveling magnetic micro-gates and edge-deployed ML (YOLO-class), implementing a latency-aware control loop for 3-way assay-targeted routing.
- Built an acoustofluidic micronozzle that selectively extrudes ~10 μm cells while flushing sub-micron bacteria

Student Research Assistant 09/2021 – 07/2023
Advanced Microfluidics and Microdevices Laboratory (AMMLAB) – Abu Dhabi, United Arab Emirates
- Created smart laparoscopic forceps utilizing force and angle sensors for assessing tissue stiffness in keyhole surgeries.
- Designed an IDE device for analyzing the physicochemical properties of RBCs in diabetic patients

Summer Intern 06/2022 – 08/2022
European Organization for Nuclear Research (CERN) – Geneva, Switzerland
- Aided in designing, assembling, and optimizing a navigation robot to inspect CMS caverns for structural failure.
- Calibrated a camera for capturing cracks and crevices in CERN caverns using Python scripts on a Raspberry Pi.
- Run analytical tests on the CERN's COVID-19 Airborne Risk Assessment (CARA) tool using similar viral infectivity models.

Skills

Experimental, Microfabrication & Cleanroom Skills:

Photolithography & resist processing (Karl Suss MABA6 Gen 4 mask aligner, Fusion DUV resist modification, spin coater, L-Edit layout design); Thin-film deposition (Temescal FC-2000 evaporator, Kurt J Lesker CMS-18 sputter); Plasma & surface treatment (Harrick Plasma O_2 cleaner, SAMCO AQ-2000); 3D micro/nanofabrication (Nanoscribe, resin 3D printer); Sample prep & materials characterization (brightfield microscopy, LatticeAx 420 cleaving station, Rame-Hart contact angle goniometer).

Computational & Modeling Skills:
Python (NumPy, pandas, scikit-learn, Jupyter; data mining & CNN-based modeling); MATLAB; COMSOL Multiphysics (FEA & multiphysics modeling); C++; CAD & simulation (SOLIDWORKS, Fusion 360, ANSYS); data analysis & statistics (JMP, STATISTICA); PCB design; embedded systems (Arduino, Raspberry Pi, microcontrollers); version control (Git, GitHub, GitLab)

Awards, Fellowships, and Grants

2026 College of Engineering Diefenderfer Graduate Fellowship in Entrepreneurship
2026 Penn State Center for Biodevices 'Ignite' Seed Grant for Medical Innovation ($60,000)
2025 Vice Provost and Dean of the Graduate School Student Persistence Scholarship
2025 1st Place — LPE 497: *Patent Essentials for Engineers & Innovators* IP Strategy Competition (PSU × Harrity & Harrity, LLP)
2025 NSF Innovation Corps Grant for Customer Discovery in Research Translation ($55,000)
2024 Society of Tribologists and Lubrication Engineers (Academic and Research Excellence) Scholarship Award
2023 IEEE Engineering in Medicine and Biology Conference (EMBC) Travel Grant (Advancement Opportunity)
2023 New England Science Symposium Travel Grant
2021 Bronze Medal at the International Genetically Engineered Machine (iGEM) competition
2021 NYU Tandon Summer Undergraduate Research Program (SURP) Grant
2015 J.H. Yorke's Best Incoming Student Award
2014 President's Independence Day Best Student Award

Publications and Presentations

2026 **Ma-sum Abdul-Hamid**, Bayan Orynbassarova, Eric Gordon, Wentian Liao, Amir Sheikhi, Igor S. Aronson. Simulation-Guided Design of a Silicon–Glass Acoustofluidic Nozzle for Size-Selective Retention. IEEE Nanotechnology Materials and Devices Conference (*IEEE NMDC 2026 - accepted for IEEE Xplore publication*).

2026 **Ma-sum Abdul-Hamid**, Igor S. Aranson. Deep Reinforcement Learning for Mode-Switching Cargo Transport by Chiral Magnetic Micropropellers. IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics; August 2026; Edmonton, Canada. ***Link***

2026 **Ma-sum Abdul-Hamid,** Bayan Orynbassarova, Igor Aronson. A Modular Computational Framework for Designing Size-Selective Acoustofluidic Nozzle Devices. Poster presented at 2026 Biomedical Innovation Exchange: Center for Biodevices Symposium & Inaugural Biomedical Engineering Research Forum; March 2026; State College, PA, USA.

2025 Alexandra Mystikou, David Nelson, Diana C. El Assal, Ashish K. Jaiswal, Mehar Sultana, Cecilia Rad-Menendez, Noura Al-Mansoori, Sewar T. Elias, **Ma-sum Abdul-Hamid**, Layanne Nayfeh, Nizar Drou, John A. Burt, David H. Green, Kourosh Salehi-Ashtiani. Earth-Observation and Environmental Vision Transformers Reveal Genome–Environment Associations in Macroalgae. *Cell Press (Multi-Journal Submission - under review).* ***Preprint***

2025 **Ma-sum Abdul- Hamid**. Investigating Z-Axis Clustering of Fibroblast Cells in Hydrogels Under Combined Magnetic and Acoustic Fields. Poster presented at GRC Conference on Complex Active and Adaptive Material Systems; January 2025; Ventura, USA.

2025 Wael Othman, Kojo E. Vandyck, **Ma-sum Abdul-Hamid**, Carlos Abril, Juan S. Barajas-Gamboa, Juan P.Pantoja, Mohammad A. Qasaimeh. Off-The-Jaw Tactile Sensing System for Tissue Stiffness and Thickness Assessment in Minimally Invasive Surgery. *IEEE Access* ***Link***

2024 Dima Samer Ali, Samuel O. Sofela, Muhammedin Deliorman, Pavithra Sukumar, **Ma-sum Abdul-Hamid**, Sherifa Yakubu, Ciara Rooney, Ryan Garrod, Anoop Menacherya, Rabih Hijazi, Hussein Saadi, Mohammad A. Qasaimeh. OMEF Biochip for Evaluating Red Blood Cell Deformability Using Dielectrophoresis as a Diagnostic Tool for Type 2 Diabetes Mellitus. *Lab on a Chip* ***Link***

2024 Sanjidah Ahmed Ira, Brooke Elise Sherry, Ruhana Amin, **Ma-sum Abdul-Hamid**, Timothy Roberts, Sarah Hochman, Cassandra L Thiel. Do reusables pose greater infectious risks than disposables for consumer goods?: A systematic literature review. *International Journal of Infection Control Vol 20* ***Link***

2023 **MA Hamid**. Incorporation of force and angle sensors in laparoscopic surgical tools. Poster presented at 2023 New England Science Symposium (Harvard Medical School) Session; April 2023; Boston, Massachusetts, USA.

2023 Othman, W., Vandyck, K. E., **Abdul-Hamid, M**., & Qasaimeh, M. A. (2023). Off-the-Jaw Force Sensing and Tissue Stiffness Assessment using a Smart and Ergonomic Laparoscopic Grasper System. Poster presented at IEEE Engineering in Medicine in Biology Conference Proceedings; July 2023; Sydney, Australia.

2022 **MA Hamid**. Analytical optimization of CERN's CARA tool. Poster presented at 2022 CERN Summer Student Presentation Session; July 2022; Geneva, Switzerland. ***Link***

Volunteer/Extracurricular Experience

2026 Graduate School Applications Mentor for the Pan-African Professional Alliance at Penn State
2025 Graduate Student Recruitment and Undergraduate Poster Presentation Judge at the 2025 ABRCMS Conference
2024 Judge at the Penn State Undergraduate Exhibition for Research, Inquiry, and Creative Activity
2023 Peer Reviewer for the International Journal of Infection Control (IJIC)
2021 Mentor for College Admissions Mentors for Africa (CAMA)

2019 Men's Intercollegiate Soccer (Musbah FC - NYU Abu Dhabi)
2018 Staff Member for the Youth Initiative for Africa's Development (YIFAD)

Certificates and Professional Memberships

Memberships: Tau Beta Pi (Engineering Honor Society), National Society of Black Engineers (NSBE), Biomedical Science Careers Program (BSCP), IEEE Engineering in Medicine and Biology Society (EMBS), American Society of Mechanical Engineers (ASME).

Course Certificates: NIH Entrepreneurship Bootcamp, Coursera (Supervised Machine Learning, Machine learning in Healthcare, Comparative Health Systems, Design and Conduct of Clinical trials, Introduction to Generative AI, Machine learning with Python).